

FPGA-based Implementation of Ultrasonic Energy Mapping for Non-destructive Evaluation Applications

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Introduction

Ultrasonic energy mapping (UEM) is very useful technique for measureable assessment of damages in the structures. It involve two dimensional Fourier transform and wavenumber filtering of a full wavefield data. To use UEM method in real field applications, we want to integrate this technique in ultrasonic propagation imaging (UPI) system. In this paper, we present the architecture of ultrasonic energy mapping algorithm that is integrated in field programmable gate array (FPGA) based ultrasonic propagation imaging system. UEM architecture is designed to perform multidimensional Fourier transform using row-column decomposition method. A source removal technique is added which is used to highlight the damage in the presence of high energy at the source. We used a composite wing box specimen to inspect and generate UEM results. Our experiment showed that we successfully detect a damage of 20 x 20 mm² in the presence of source inside the scan area. The system only takes less than 5 seconds to process around 57,600 points, which is very practical for real-world applications.

Ultrasonic Energy Mapping in FPGA

Ultrasonic energy mapping algorithm takes the full wavefield data that is acquired using ultrasonic propagation imaging (UPI) system [1] and perform two dimensional Fourier transform, wavenumber filtering, and wave energy calculation of each scan point at the scan area. These computations are performed in FPGA, where multi-dimensional Fourier transform is implemented using row-column decomposition method. Row-column decomposition method is very suitable for parallel implementation of multi-dimensional Fourier transform and it reduces the complex multiplications by a large factor [2]. This method decomposes the 2D Fourier transform computation into multiple 1D transformations, first along row and then along columns. We used Xilinx DFT/FFT IP cores to compute 1D Fourier transform in FPGA. A source removal technique is also added which uses a mask to block the high source energy found at the sensor location. Signals induced from the damages are much smaller in energy than the source, therefore source removal technique is applied to highlight the damages. Mathematically, UEM is presented using equations (1), (2), and (3) as follows.

$$W_n(k_x, k_y, t) = F_{2D}\{w(x, y, t)\} \quad (1)$$

$$\tilde{w}_n(x, y, t) = F_{2D}^{-1}\{\tilde{W}_n(k_x, k_y, t)\} \quad (2)$$

$$E(x, y) = \int_0^T \frac{1}{2} \{\tilde{w}_n(x, y, t)\}^2 dt \quad (3)$$

where, $w(x, y, t)$ denotes the full wavefield time domain data and its two dimensional Fourier transformation is denoted as $W_n(k_x, k_y, t)$. A wavenumber filter is applied to $W_n(k_x, k_y, t)$, which gives us narrow-band wavefield in the wavenumber domain $\tilde{W}_n(k_x, k_y, t)$. An inverse Fourier transform is then applied and wave energy $E(x, y)$ is calculated at each grid point as shown in equation (3).

Results

Experiment was performed on a composite wing box specimen. A scan area of $240 \times 240 \text{ mm}^2$ was inspected and 0.25 mm central wavenumber was applied to generate UEM plot. The sensor was located inside the scan area as shown in Fig. 1(a). The processing time results obtained for various inspection area sizes showed that it only take less than 5 seconds to inspect a scan area of 240×240 points. Figure 1(c) and (d) showed the contrast between two results of UEM that are generated with and without source removal technique. The result showed that impact damage of size $20 \times 20 \text{ mm}^2$ was more visible when source removal mask was applied to the scan area.

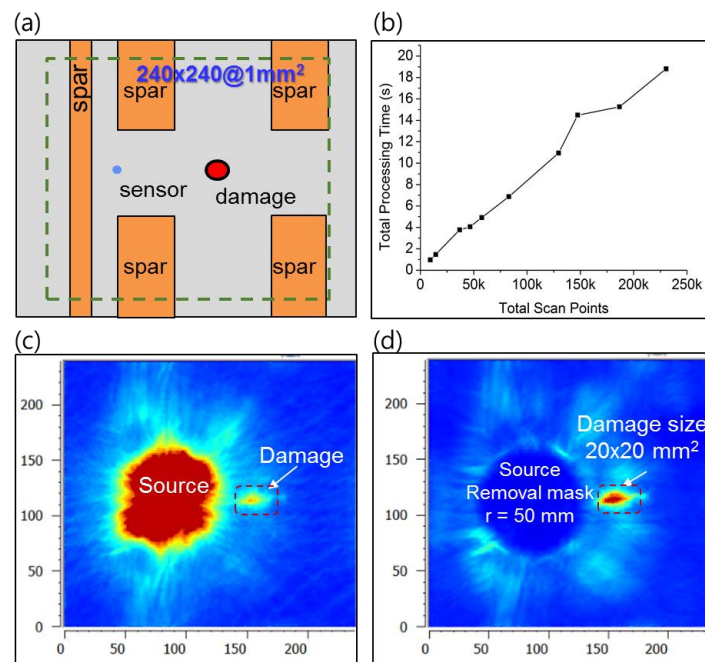


Figure 1: a) location of inspection area on a wing box specimen, b) performance graph of FPGA-based UEM, c) FPGA-based UEM without applying source removal, d) FPGA-based UEM using source removal.

Conclusion

UEM technique is implemented in FPGA based UPI system. Source removal method is also incorporated with UEM technique and due to which the system is able to detect a damage in presence of high source energy. The results showed that the wavefield time domain data is 57,600 points is processed in less than 5 seconds to generated UEM result.

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